

A disproof of Conjecture 00000002591

No finite poset with at least three elements has $2^{|P|}$ Dedekind–MacNeille cuts

1 The conjecture

Conjecture 00000002591 reads:

Definition: The Dedekind–MacNeille completion $\text{DM}(P)$ is the smallest complete extension of a poset P preserving all existing joins and meets. *Conjecture:* The size law of MacNeille completions: $|\text{DM}(P)| \leq 2^{|P|}$ with the upper bound attained by unbounded dense antichains; the isomorphism types of P attaining the bound are superpolynomial in number, with the minimal generator characterization by three-chain cross products.

It is a conjunction of four clauses:

- (i) $|\text{DM}(P)| \leq 2^{|P|}$;
- (ii) the bound is attained by (unbounded dense) antichains;
- (iii) the number $E(n)$ of isomorphism types of n -element posets P with $|\text{DM}(P)| = 2^n$ is superpolynomial in n ;
- (iv) a “minimal generator characterization by three-chain cross products”.

We prove (i) for finite posets (the Lean theorem `card_cuts_le` assumes `Fintype`; the same injection of cuts into subsets gives the inequality for arbitrary cardinals, but that general statement is not formalized), and we refute (iii) and (ii) for finite posets: $E(n) = 0$ for every $n \geq 3$. Clause (iv) is too vague to formalize and is not addressed; refuting (iii) already refutes the conjunction.

Readings of (iii) that are refuted. $E(n)$ is counted either up to isomorphism or as labelled partial orders on $\{0, \dots, n-1\}$; “attaining the bound” is read either as $|\text{DM}(P)| = 2^n$ or as $|\text{DM}(P)| \geq 2^n$ (by (i) these are the same); “superpolynomial” is read either strongly (for every k , $E(n) > n^k$ for all large n) or weakly (for every k and C , $E(n) > Cn^k$ for infinitely many n , i.e. E is not $O(n^k)$ for any k). All of these are refuted.

Not refuted / not formalized. Clause (ii) for infinite posets is not formalized. (In prose: an infinite antichain of cardinality κ has $\kappa + 2 = \kappa < 2^\kappa$ cuts, so infinite antichains do not attain the bound either; infinite dense linear orders such as $\mathbb{Q}$ do attain $2^{\aleph_0}$, but they are not antichains.) Clause (iv) is not addressed.

2 Definitions and conventions

For $A \subseteq P$ let A^u be the set of upper bounds of A and A^ℓ the set of lower bounds. Following the retrieved Wikipedia article on the Dedekind–MacNeille completion (https://en.wikipedia.org/wiki/Dedekind%E2%80%93MacNeille_completion), $\text{DM}(P)$ “consists of all subsets A for which $(A^u)^\ell = A$, ordered by inclusion”; equivalently a *cut* is “a pair of sets (A, B) for which $A^u = B$ and $A = B^\ell$ ”.

In Lean we use Mathlib’s `DedekindCut P`, which is `Concept P P (. <= .)`: pairs `(left, right)` with `upperBounds left = right` and `lowerBounds right = left`. This is exactly the pair form of a cut; a cut is determined by its left set (`Concept.extent_injective`), and a set s is a left set iff `lowerBounds(upperBounds s) = s` (Lean: `exists_cut_iff`). So $|\text{DM}(P)|$ is `Nat.card (DedekindCut P)`.

Empty-set convention. No extra elements are adjoined. The empty set is a left set iff P has no least element; P itself is always a left set. For example, an antichain with $n \geq 2$ elements has the $n + 2$ cuts $\emptyset$, the n singletons and P , matching the Wikipedia description of the completion of an antichain as “ S itself together with two additional elements”. The argument below uses only subsets with one or two elements, so the conclusion does not change if $\emptyset$ and P are always counted as cuts.

Counting. Isomorphism types of n -element posets are represented by partial orders on `Fin n`. `Attaining n` is the type of partial orders Q on `Fin n` with $2^n \leq |\text{DedekindCut}|$ for Q , and $E(n)$ is the cardinality of its quotient by order isomorphism: `E n = Nat.card (Quot Iso)`. Both types are finite (Lean instances), so $E(n)$ is a genuine count.

3 Proof

Lemma 1 (`no_three_of_cuts`). *If every singleton and every two-element subset of P is the left set of a cut, then P has no three pairwise distinct elements.*

Proof. For $u \in P$, $\{u\}^u = \uparrow u$ and $(\uparrow u)^\ell = \downarrow u$, so $\{u\}$ being closed gives $\downarrow u = \{u\}$ for all u . Let x, y, z be pairwise distinct. If u is an upper bound of $\{x, y\}$ then $x, y \in \downarrow u = \{u\}$, so $x = u = y$, a contradiction; hence $\{x, y\}^u = \emptyset$ and $(\{x, y\}^u)^\ell = \emptyset^\ell = P$. Since $\{x, y\}$ is closed, $P = \{x, y\}$, so $z \in \{x, y\}$, a contradiction. $\square$

Lemma 2 (`exists_proper_not_cut`). *If $|P| \geq 3$, some nonempty proper subset of P (of size one or two) is not the left set of a cut.*

Proof. Pick three distinct elements. If some singleton $\{u\}$ is not a left set, it is nonempty and proper (it misses one of the other elements). Otherwise, by the previous lemma some pair $\{u, v\}$ is not a left set; it is proper because it cannot contain three distinct elements. $\square$

Theorem 3 (`card_cuts_le`, `card_cuts_lt`). *For a finite poset P , $|\text{DM}(P)| \leq 2^{|P|}$; if $|P| \geq 3$ then $|\text{DM}(P)| < 2^{|P|}$.*

Proof. The map from cuts to their left sets is injective into the power set, which has $2^{|P|}$ elements. If $|P| \geq 3$, the previous lemma gives a subset outside its range, so the inequality is strict. $\square$

In particular no antichain with at least three elements attains the bound, which refutes clause (ii) for finite antichains.

Theorem 4 (`not_superpolynomial`). *$E(n) = 0$ for all $n \geq 3$. Consequently it is false that for every k , $E(n) > n^k$ eventually, and it is false that for every k, C , $E(n) > Cn^k$ infinitely often.*

Proof. By the previous theorem, `Attaining n` is empty for $n \geq 3$ (`attaining_isEmpty`), so both the labelled count (`card_attaining_eq_zero`) and $E(n)$ are 0. Taking $k = 0$ (and $C = 0$), either growth statement would give some $n \geq 3$ with $E(n) > 0$. $\square$

Remark 5 (non-vacuity, `antichain2_attains`, `E_two_pos`). The bound is sharp at $n = 2$: the two-element antichain on `Fin 2` has all four subsets as left sets of cuts, so $E(2) > 0$. This checks that the definitions of `Attaining` and E are not vacuous. A brute-force check also confirms that no poset on 3 or 4 points has 2^n cuts and that antichains on 3 and 4 points have $n + 2$ cuts.

4 Lean formalization

File `lean/Conjecture2591/Basic.lean` (namespace `Conjecture2591`, 206 lines, only import `Mathlib`). Main theorem (ASCII rendering):

```
theorem not_superpolynomial :
  (forall n, 3 <= n -> E n = 0) /\
  not (forall k : Nat, forall^f n in atTop, n ^ k < E n) /\
  not (forall k C : Nat, exists^f n in atTop, C * n ^ k < E n)
```

(`forall^f` and `exists^f` stand for Lean's `Filter.Eventually` and `Filter.Frequently`.) Supporting results: `exists_cut_iff`, `no_three_of_cuts`, `exists_proper_not_cut`, `card_cuts_le`, `card_cuts_lt`, `attaining_isEmpty`, `card_attaining_eq_zero`, `E_eq_zero`, `antichain2_attains`, `E_two_pos`.

Axiom audit. `#print axioms` for `not_superpolynomial`, `card_cuts_lt` and `E_two_pos` reports only `propext`, `Classical.choice`, `Quot.sound`. There is no `sorry` and no `native_decide`.